Week 6

1) The key-value abstraction is best described as:
A. A directed graph of objects
B. A log-structured file system
C. A relational schema with joins
D. A dictionary data structure mapping keys to values

2) A key reason NoSQL gained popularity for modern workloads is that:
A. It requires stronger schemas than SQL
B. Modern data is often large and unstructured with many random reads/writes
C. It emphasizes foreign keys and joins
D. It runs only on a single machine

3) Column-oriented storage is useful mainly because:
A. It makes joins faster
B. It eliminates indexes
C. It allows faster range searches within a column without fetching all columns
D. It guarantees linearizability

4) Cassandra maps keys to servers using a component called the:
A. Partitioner
B. Coordinator
C. Snitch
D. Bloom Filter

5) The “Hinted Handoff” mechanism ensures Cassandra is “always writable” by:
A. Rejecting writes when any replica is down
B. Buffering writes and later delivering them to recovered replicas
C. Forcing all writes to go to ALL replicas
D. Disabling replication temporarily

6) Cassandra’s SSTables are:
A. Mutable files updated in place
B. Used only for caching
C. Only indexes without data
D. Immutable files sorted by key

7) A Bloom filter in Cassandra guarantees:
A. No false positives and no false negatives
B. Possible false negatives but no false positives
C. Possible false positives but no false negatives
D. Always exact membership check

8) In modern cloud systems where partitions must be tolerated, CAP implies you must choose between:
A. Latency and throughput
B. Consistency and availability
C. Storage and compute
D. Encryption and compression

9) Cassandra is presented as prioritizing:
A. Strong consistency and availability, but not partition tolerance
B. Strong consistency and partition tolerance, but not availability
C. Eventual consistency, availability, and partition tolerance
D. Only consistency (ACID)

10) “Eventual consistency” means that if writes stop for a key:
A. Replicas diverge permanently
B. All replicas eventually converge to the same value
C. Reads always return the latest value immediately
D. The key becomes unavailable

11) LOCAL_QUORUM differs from QUORUM because LOCAL_QUORUM:
A. Requires all DCs to respond
B. Requires only one replica
C. Waits for quorum only within the coordinator’s datacenter
D. Disables read repair

12) Causal consistency requires that reads:
A. Always return the latest write in real time
B. Respect partial order based on information flow
C. Execute in a single global total order
D. Never return stale values

13) In HBase, if a timestamp is not specified during a read:
A. The oldest version is returned
B. The operation fails
C. The latest version is returned
D. A random version is returned

14) HBase ensures strong write durability using:
A. Gossip membership
B. Write-ahead log (HLog) before MemStore updates
C. Hinted handoff
D. Quorum reads only